Design of a Programmable Gain Amplifier Using a Fully Differential OTA with 5T-OTA CMFB and DC Offset Cancellation

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1. Introduction and Specifications

This report covers a programmable gain amplifier (PGA) built around a two-stage fully differential Miller OTA, with a 5T-OTA continuous-time CMFB and an RC-integrator DC offset canceller. The design is in SCL 180 nm CMOS at 1.8 V with a 1 pF load, and all results come from Cadence Virtuoso simulation.

Specifications

Parameter	Target
DC open-loop gain	⩾ 70 dB
Open-loop −3 dB bandwidth	⩾ 75 kHz
Phase margin	60°–80°
CMRR	⩾ 80 dB
PSRR	⩾ 65 dB
Slew rate	⩾ 20 V/µs
Settling time (2%)	⩽ 200 ns
Power (OTA + CMFB)	⩽ 1 mW
Input-referred noise at 1 kHz	⩽ 35 nV/√Hz
Output offset after DCOC	⩽ 10 mV
PGA gain range	10–50 dB
PGA −3 dB bandwidth at 50 dB	⩾ 750 kHz
PGA gain error	⩽ 5%
Total power including DCOC	< 1.5 mW
CMFB loop gain / UGF / PM	⩾ 40 dB / 5–10 MHz / ⩾ 60°

Note on the bandwidth targets. An open-loop −3 dB point of 75 kHz together with 70 dB DC gain implies a unity-gain bandwidth near 237 MHz. A 750 kHz bandwidth at 50 dB gain (316 V/V) implies the same figure. Section 3 shows this needs roughly 4–5 mW, so the two bandwidth targets conflict with the 1 mW budget. The design is sized for the power budget, and Section 8 gives the resulting figures.

2. Architecture and Process

- **Core OTA:** two-stage, fully differential, Miller Rc–Cc compensation. A folded cascode would stack 5–6 overdrives and cannot give the required 1.4 V swing at 1.8 V; the two-stage needs only four.
- **CMFB:** 5T-OTA senses the output common mode through two 500 kΩ resistors and compares it with Vref = 0.9 V. The error drives the second-stage current sources.
- **Gain network:** a switched resistor ladder. NMOS switches short ladder segments, so gain depends on resistor ratios and tracks across corners.
- **DCOC:** an RC integrator in feedback acts as a high-pass at fHP ≈ 2 kHz, suppressing DC and near-DC offset.

Extracted process parameters

Parameter	Value	Note
Lmin / Wmin	180 nm / 220 nm	
Vth,N / Vth,P	477.2 mV / 513.7 mV	VSB = 0
μnCox / μpCox	234.5 / 63.4 μA/V²	
ro,N / ro,P	149.8 kΩ / 108.1 kΩ	125 μA, L = 180 nm
Cox	≈ 8.6 fF/μm²	estimated

3. Hand Calculations: Two-Stage OTA

The OTA is sized for UGB = 53 MHz, the most the 1 mW budget allows. Each step below gives the expression, the substitution and the result. Hand values are estimates and are compared with simulation in Section 7.

Assumptions: Cc = 3 pF, Vov1 = 150 mV, CL = 1 pF, L = 500 nm for gain-critical devices.

3.1 Required transconductance (4a)

Quantity	Expression	Substitution	Result
UGB for 750 kHz at 50 dB	Gmax × BW	316 × 750 kHz	237 MHz
gm1 for that UGB	2π × UGB × Cc	2π × 237 MHz × 3 pF	4.47 mA/V
Tail current for that gm1	gm1 × Vov1	4.47 mA/V × 150 mV	670 µA (4–5 mW total)
UGB chosen (power-limited)	—	—	53 MHz
gm1 chosen	2π × UGB × Cc	2π × 53 MHz × 3 pF	1.00 mA/V

The 237 MHz target costs more than 4× the power budget, so 53 MHz is used.

3.2 Input pair bias and sizing (4b)

Quantity	Expression	Substitution	Result
ID1	gm1 × Vov1 / 2	1 mA/V × 0.150 V / 2	75 µA
Itail	2 × ID1	2 × 75 µA	150 µA
Slew rate	Itail / Cc	150 µA / 3 pF	50 V/µs
(W/L)1	$gm1^2$ / (2 µnCox ID1)	$(1\ mS)^2$ / (2 × 234.5 µ × 75 µ)	28.4
W1	(W/L)1 × L	28.4 × 500 nm	14 µm

3.3 Open-loop gain, poles and compensation (4c)

Quantity	Expression	Substitution	Result
Rout1	ro1 ‖ ro3	694 kΩ ‖ 360 kΩ	237 kΩ
Rout2	ro5 ‖ ro7	469 kΩ ‖ 650 kΩ	272 kΩ
A1	gm1 × Rout1	1 mS × 237 kΩ	237 V/V (47.5 dB)
A2	gm5 × Rout2	0.8 mS × 272 kΩ	218 V/V (46.8 dB)
A0	A1 × A2	237 × 218	51,700 V/V (94 dB)

Quantity	Expression	Substitution	Result
Dominant pole fp1	1 / (2π gm5 Rout2 Rout1 Cc)	1 / (2π × 0.8 m × 272 k × 237 k × 3 p)	1 kHz
Second pole fp2	gm5 / (2π CL)	0.8 mS / (2π × 1 pF)	127 MHz
RHP zero (no Rc)	gm5 / (2π Cc)	0.8 mS / (2π × 3 pF)	42 MHz
Nulling resistor Rc	1 / gm5	1 / 0.9 mS	1.1 kΩ (1.5 kΩ used)
Phase margin	90° − arctan(UGB / fp2)	90° − arctan(53 / 127)	67°

Without Rc the zero lies near UGB and would destroy the phase margin. The hand gain is an upper estimate; the simulated gain is 75.4 dB.

3.4 Remaining OTA devices

PMOS devices use W/L = gm² / (2 μpCox ID). NMOS current sources use W/L = 2 ID / (μnCox Vov²).

Device	Vov (mV)	ID (μA)	gm (μS)	W/L	L (nm)	W (μm)
M3, M4 (load)	200	75	750	59	360	21
M5, M6 (second stage)	200	80	800	63	500	32
M7, M8 (output source)	150	80	—	30	500	15
M0 (tail)	200	150	—	32	500	16

Check	Expression	Result
Minimum input CM	Vov0 + Vth,N + Vov1 = 200 + 477 + 150 mV	827 mV, below the 900 mV used
Output swing	(1.8 − 0.200) − 0.150	1.45 V
Mirror pole	gm3 / (2π Cgs3) = 750 μ / (2π × 43 f)	2.8 GHz, far above UGB

L is kept at 360–500 nm because L = 180 nm gives a hand gain of only about 65 dB; a longer channel raises ro by up to 2.8×.

3.5 Switches and body effect (4f)

Quantity	Expression	Substitution	Result
Ron limit	5% of smallest segment	0.05 × 31.6 kΩ	1.58 kΩ
Switch overdrive	VDD − Vth,N	1.8 − 0.477	1.323 V
(W/L) of switch	1 / (Ron μnCox Vov)	1 / (1580 × 234.5 μ × 1.323)	2.04
Minimum W at 180 nm	(W/L) × L	2.04 × 180 nm	0.37 μm
Body effect at mid-rail	VSB ≈ 0.9 V	ΔVth	≈ 225 mV

Body effect raises Ron at mid-rail, so W = 4 μm, L = 180 nm is used. For M1/M2, VSB ≈ 0.27 V gives ΔVth ≈ 70 mV, which Vov1 = 150 mV absorbs.

3.6 Final transistor sizes

Device	Type	W (μm)	L (nm)	Role
M1, M2	NMOS	14	500	Input pair
M3, M4	PMOS	21	360	First-stage load
M5, M6	PMOS	32	500	Second-stage drivers
M7, M8	NMOS	15	500	Second-stage sources
M0	NMOS	16	500	Tail source
M9	NMOS	4	500	CMFB tail
M10, M11	NMOS	3.6	500	CMFB input pair
M12, M13	PMOS	5.3	360	CMFB load
Sw1–Sw3	NMOS	4	180	Gain switches

Passives: Cc = 3 pF, Rc = 1.5 kΩ, Rsense = 500 kΩ, CMFB node capacitor = 5 pF, Vb = 627 mV.

4. CMFB Design (4d)

The CMFB runs from a quarter of the main tail current so that it stays slower than the differential loop.

Quantity	Expression	Substitution	Result
Itail,CMFB	Itail / 4	150 µA / 4	37.5 µA
ID10	Itail,CMFB / 2	37.5 µA / 2	18.75 µA
ro,N	149.8 kΩ × (125 µ / ID) × (L / 180 nm)	149.8 k × (125 / 18.75) × (500 / 180)	2.78 MΩ
ro,P	108.1 kΩ × (125 µ / ID) × (L / 180 nm)	108.1 k × (125 / 18.75) × (360 / 180)	1.44 MΩ
Rout,CMFB	ro,N ‖ ro,P	2.78 M ‖ 1.44 M	950 kΩ
gm10 for 40 dB	100 / Rout,CMFB	100 / 950 kΩ	105 µS (minimum)
gm10 chosen	2 ID10 / Vov10	2 × 18.75 µ / 0.150	250 µS
Loop gain	gm10 × Rout,CMFB	250 µ × 950 k	237 V/V (47.5 dB)
UGF with no added cap	gm10 / (2π × 87 fF)	250 µ / (2π × 87 f)	459 MHz
Node capacitor	gm10 / (2π × 8 MHz)	250 µ / (2π × 8 M)	5 pF
UGF with 5 pF	gm10 / (2π CL,CMFB)	250 µ / (2π × 5 p)	7.96 MHz
Sense-resistor pole	1 / (2π Rsense Cgs10)	1 / (2π × 500 k × 10.3 f)	30.9 MHz

Rsense = 500 kΩ is much larger than Rout2 = 272 kΩ, and its pole is well above the loop UGF.

Sizes: M10/M11 W = 3.6 µm (L = 500 nm), M12/M13 W = 5.3 µm (L = 360 nm), M9 W = 4 µm (L = 500 nm).

The simulated CMFB loop (Section 7.3) has more gain and a higher UGF than this estimate, because the loop also includes the second-stage transconductance.

5. PGA Gain Network

The gain is set by R2/R1 with R1 = 10 kΩ at each input. The feedback ladder is built from four series segments, and NMOS switches Sw1–Sw3 short the upper segments to lower the gain.

Segment	Value
Rseg0 (always in path)	31.6 kΩ
Rseg1	114.4 kΩ
Rseg2	537 kΩ
Rseg3	2.477 MΩ

Gain code	Switches closed	R2 (effective)	Ideal gain
50 dB	none	3.16 MΩ	316 V/V
36.7 dB	Sw3	683 kΩ	68.3 V/V
23.3 dB	Sw3, Sw2	146 kΩ	14.6 V/V
10 dB	Sw3, Sw2, Sw1	31.6 kΩ	3.16 V/V

Switch Ron adds to the segment it shorts, which is why the switches are made large (4 µm / 180 nm). The gain error at each code is in Section 7.4.

6. DC Offset Cancellation (4e)

A loop around the OTA integrates the output offset and feeds it back to the input, which gives a high-pass response. The cutoff trades offset suppression against distortion of low-frequency signals.

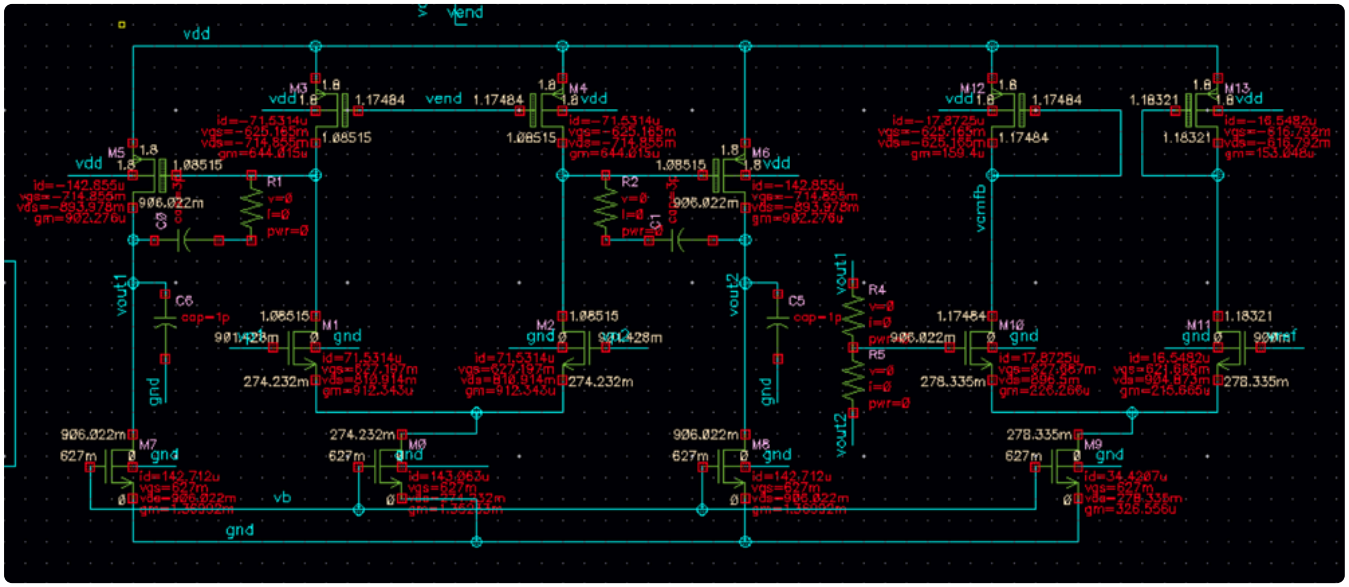
- **Cutoff:** fHP = 2 kHz is low enough to remove drift and DC offset, and high enough to leave content above 2 kHz untouched. The spec suggests 1–10 kHz.
- **Time constant:** $\tau = 1 / (2\pi f_{HP}) = 1 / (2\pi \times 2\ \text{kHz}) = 79.6\ \mu\text{s}$.
- **Components:** with COC = 10 pF (feasible on chip), ROC = τ / COC = 7.96 MΩ, so 8 MΩ is used. Then fHP = 1 / (2π × 8 MΩ × 10 pF) = 1.99 kHz.
- **Convergence:** about 5τ = 398 µs.
- **Measured offset:** the output offset with zero input and DCOC off is 5.97 mV, already inside the 10 mV limit.

7. Simulation Results

All results are for the 50 dB gain code unless stated otherwise.

7.1 DC operating point

The circuit draws 462.9 µA from 1.8 V, which is 833 µW for the OTA, CMFB and bias.



Quantity	Value
Tail M0	143.1 µA
M1/M2	71.5 µA, gm = 912 µS, Vgs = 627 mV
M5/M6	142.9 µA, gm = 902 µS,
M7/M8	142.7 µA, gm = 1.36 mS
CMFB M9 / M10, M11	34.4 µA / 17.9, 16.6 µA
Output common mode	906 mV (Vref = 900 mV)
Input common mode / tail node	901 mV / 274 mV
Bias Vb / CMFB control node	627 mV / 1.175 V

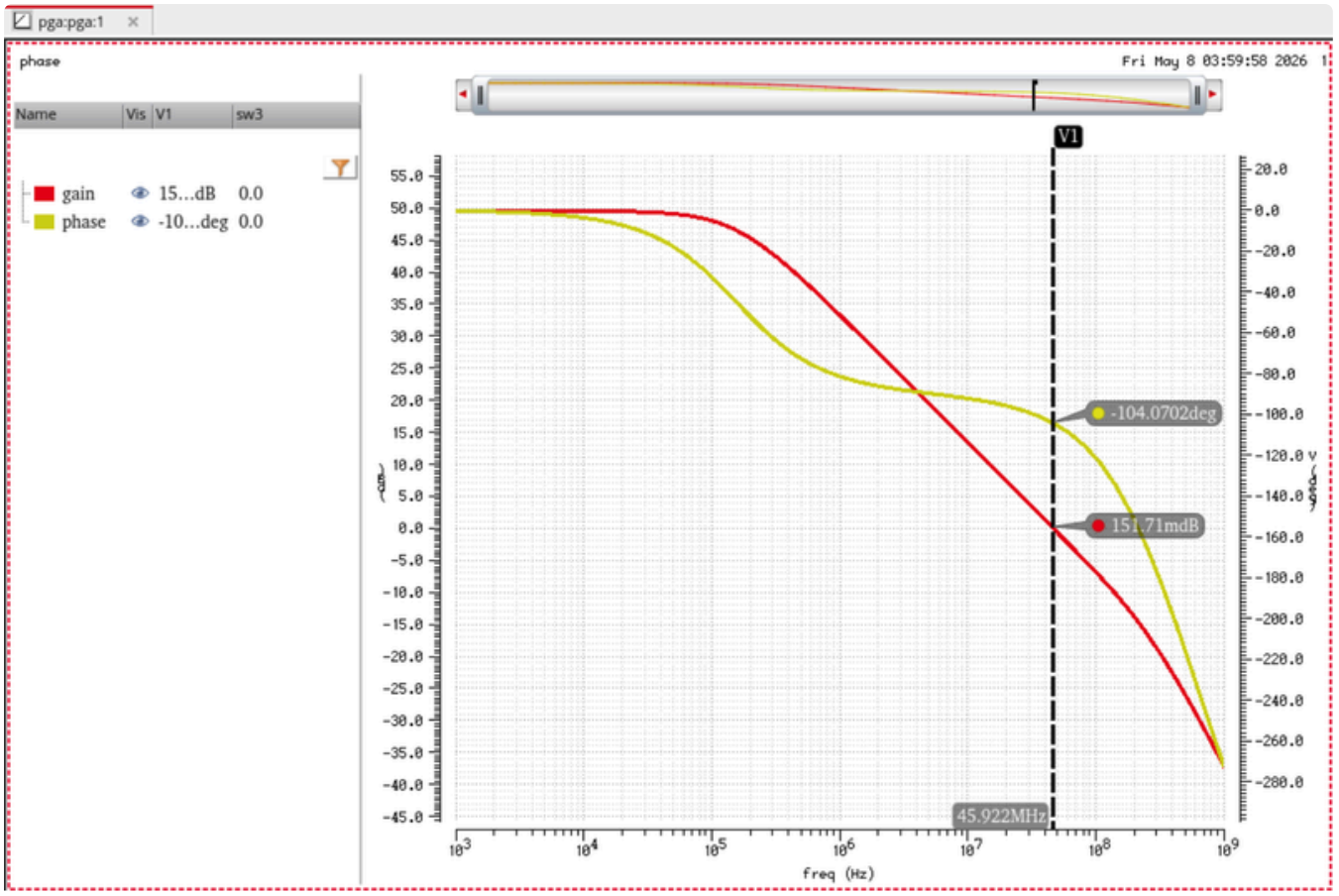
The second stage runs at 143 µA, about twice the 80 µA assumed in the hand calculation, which is why simulated gm5 (902 µS) is close to the hand value despite a lower Vov.

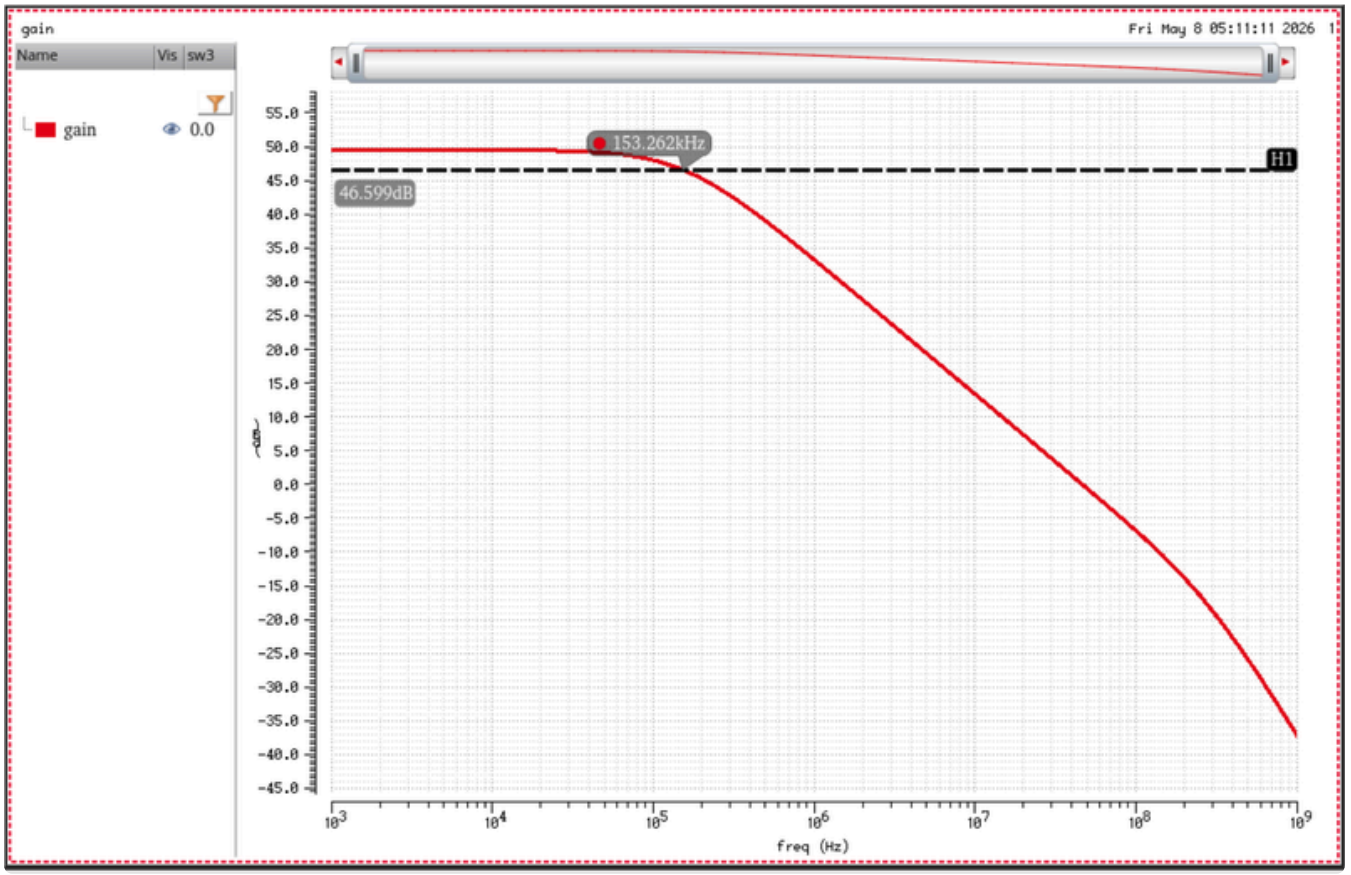
7.2 Small-signal AC

Quantity	Hand	Simulated
DC open-loop gain	94 dB	75.4 dB
Unity-gain bandwidth	53 MHz	42.5 MHz
gm1	1.00 mS	0.912 mS
Phase margin	67°	85.6°

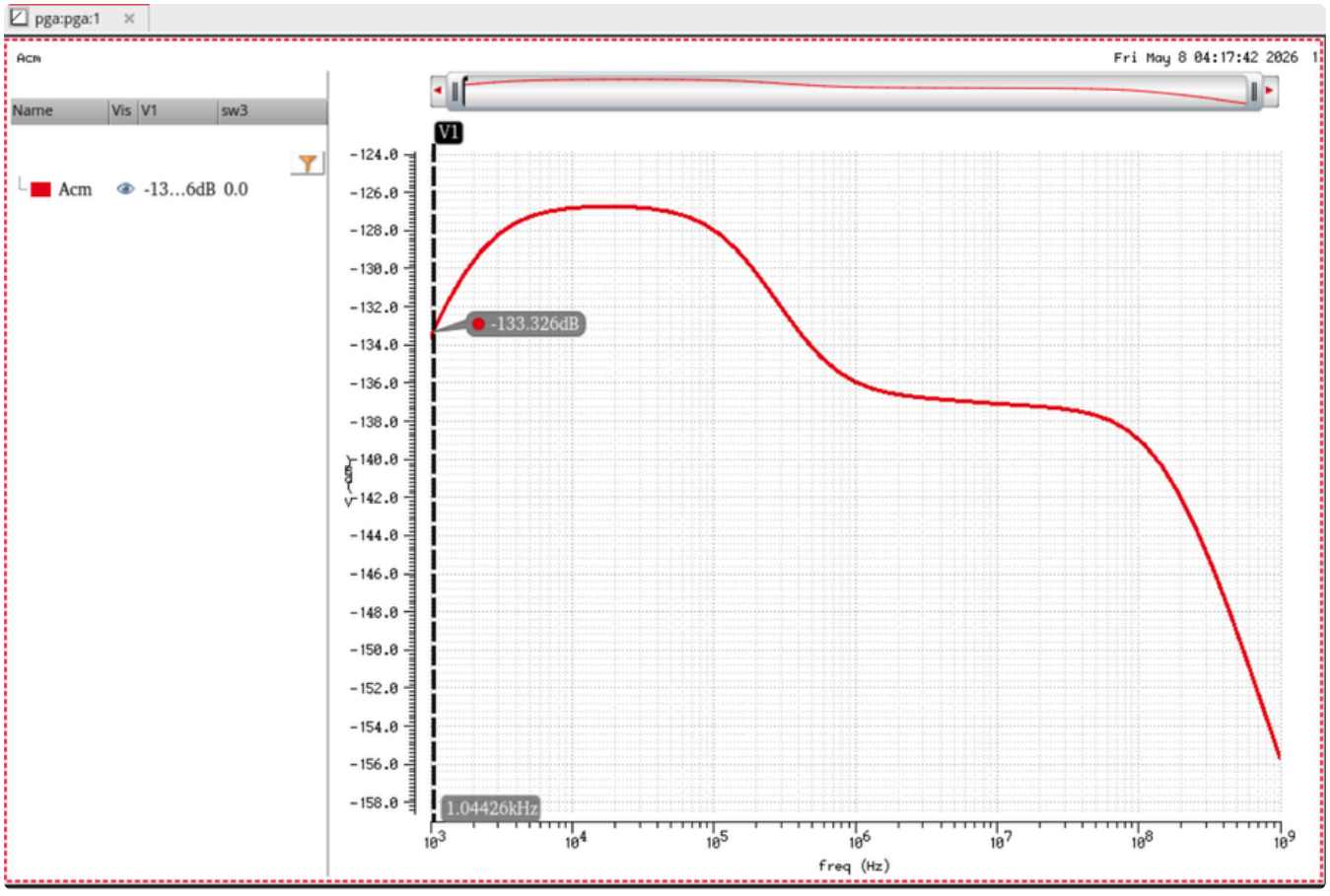
The hand gain is 19 dB above simulation. It uses ro values scaled from a single bias point and ignores the finite output resistance of the switch network and the loading of the first stage, so the simulated figure is the one to rely on. It still meets the 70 dB target.

With the 50 dB feedback code the closed-loop response has 49.5 dB gain, a −3 dB frequency of 153 kHz and a 0 dB crossing at 45.9 MHz.





Common-mode gain is shown below, and the CMRR table follows. These CMRR values come from an ideal, perfectly matched simulation, so they are upper bounds.

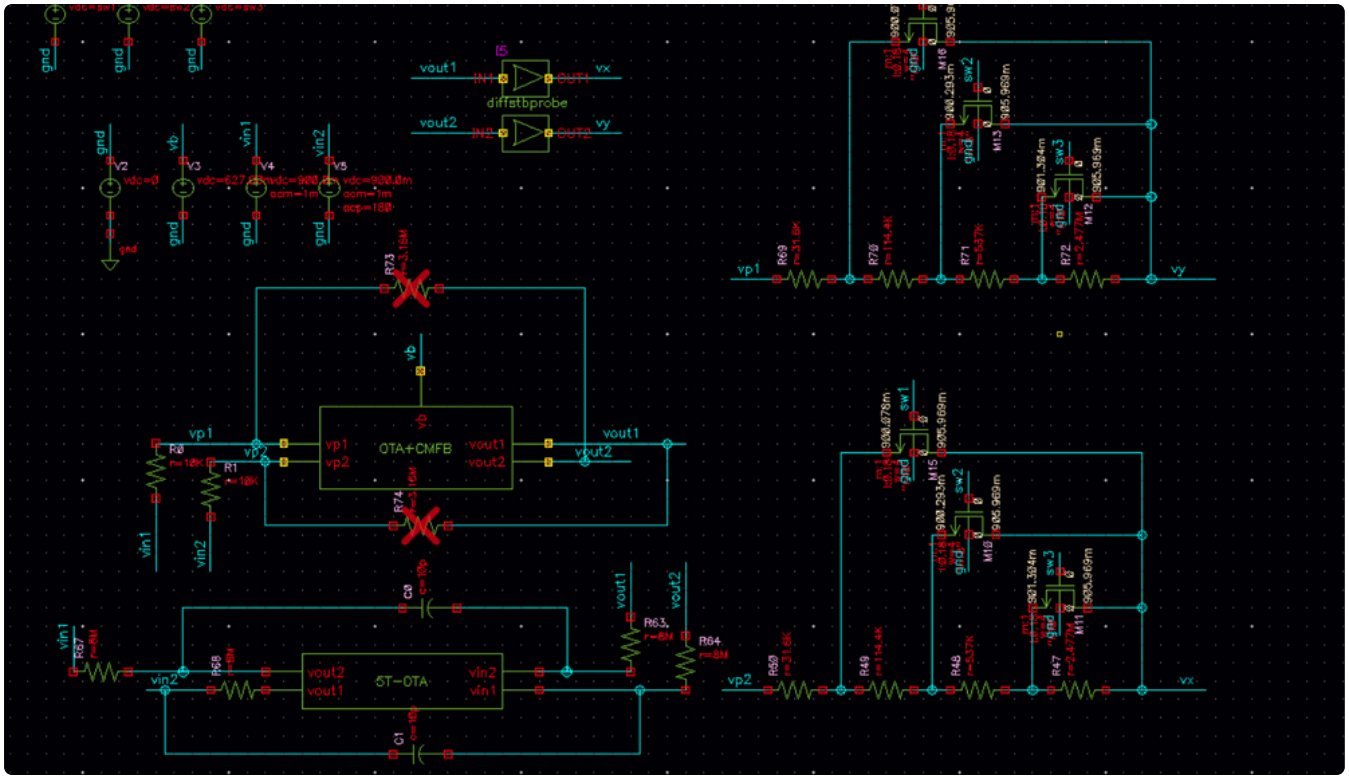


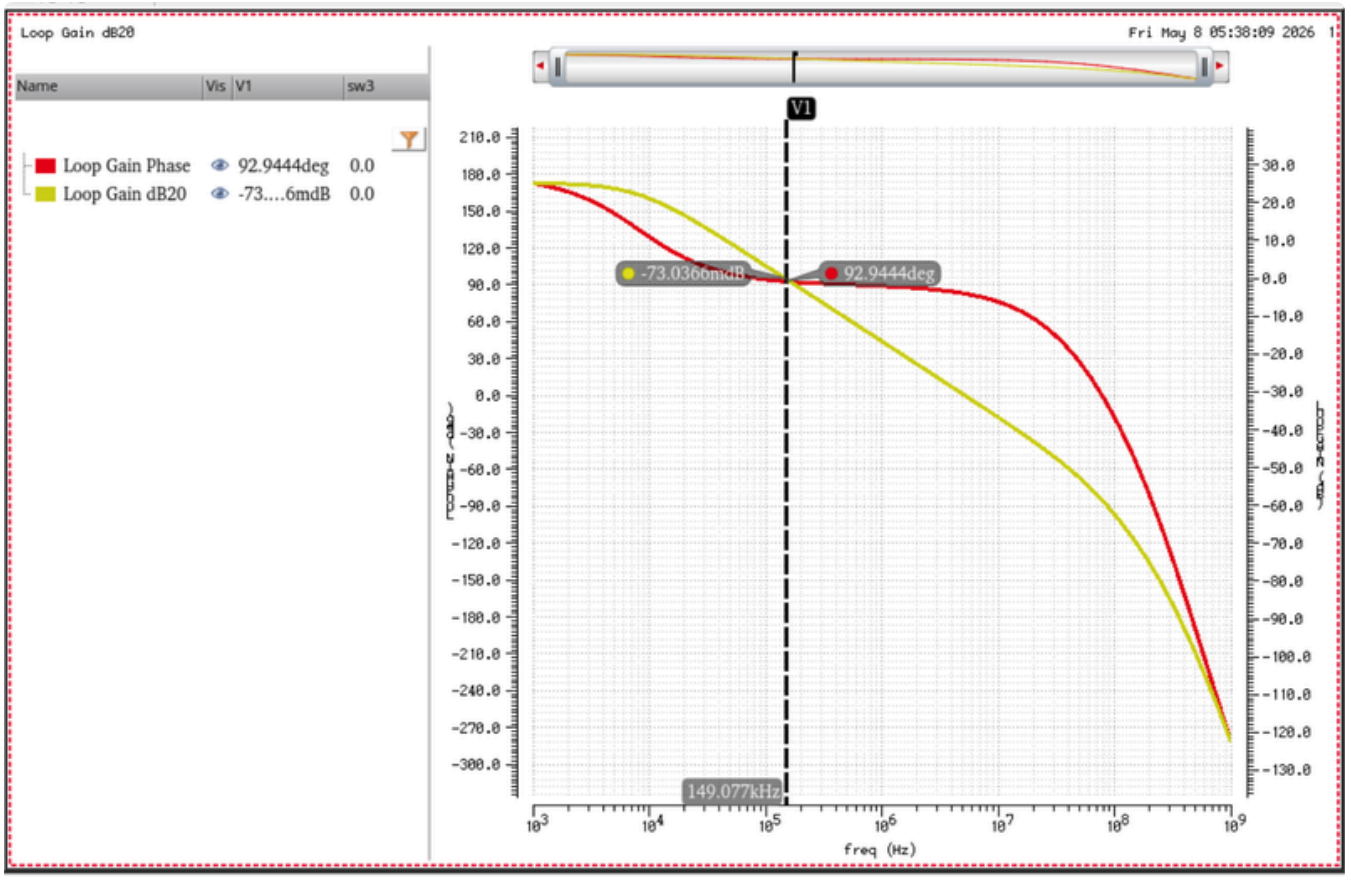
Frequency	Ad (dB)	Acm (dB)	CMRR (dB)
10 kHz	49.5	−126.8	176.3
1 MHz	33.3	−135.9	169.2
10 MHz	13.5	−137.1	150.5

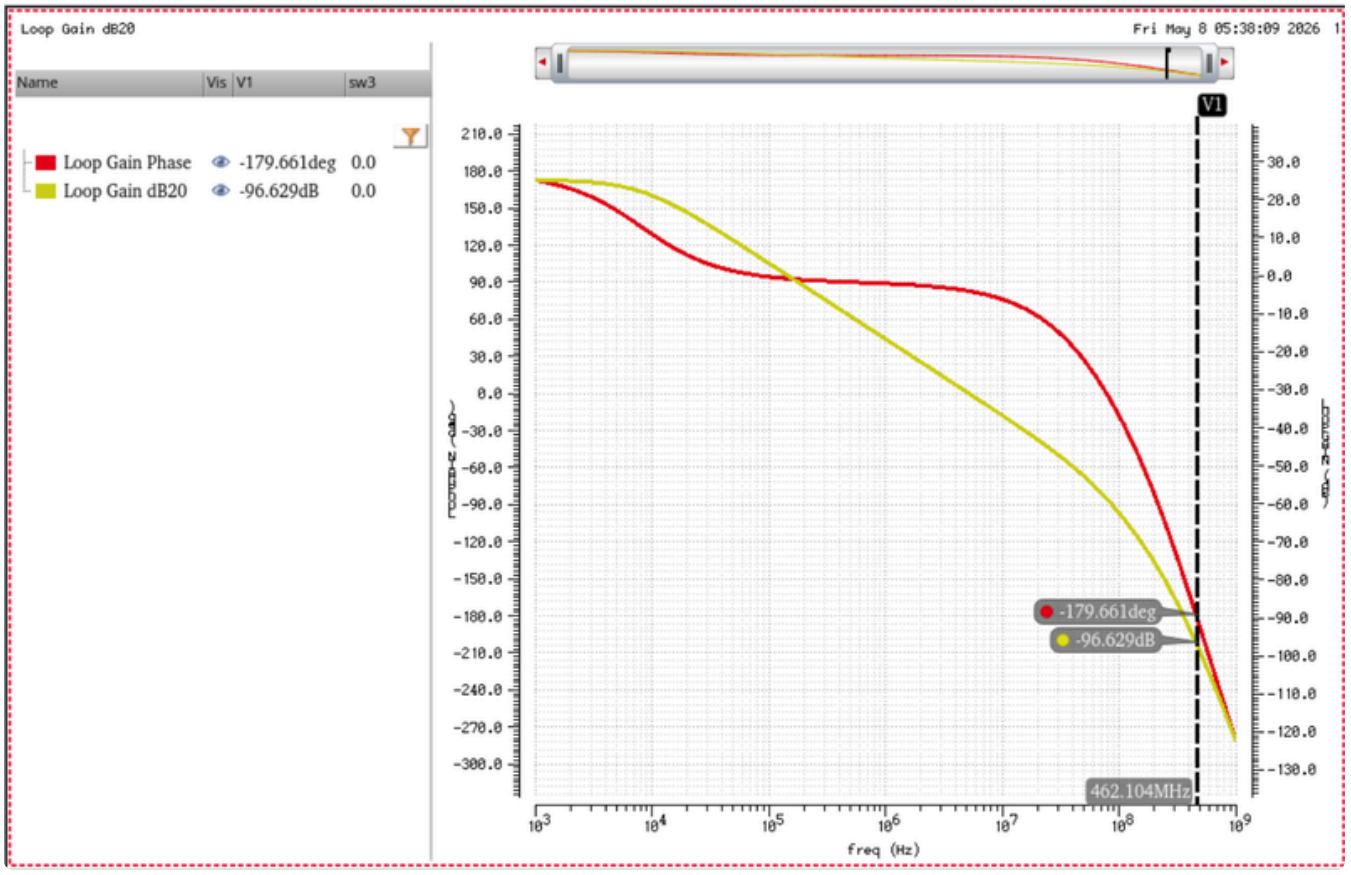
CMRR falls with frequency because tail-node parasitic capacitance lowers the tail impedance.

7.3 Stability

The differential loop was broken at the marked points in the top-level schematic and analysed with stb.


gnd
vout1
vout2
diffstbprobe
vx
vy
vb
vin1
vin2
OTA+CMFB
vp1
vp2
vout1
vout2
5T-OTA
R73
R74
C0
C1
R63
R64
R67
R68
R69
R70
R71
R72
R59
R49
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M12
M15
M14
M11
vp2

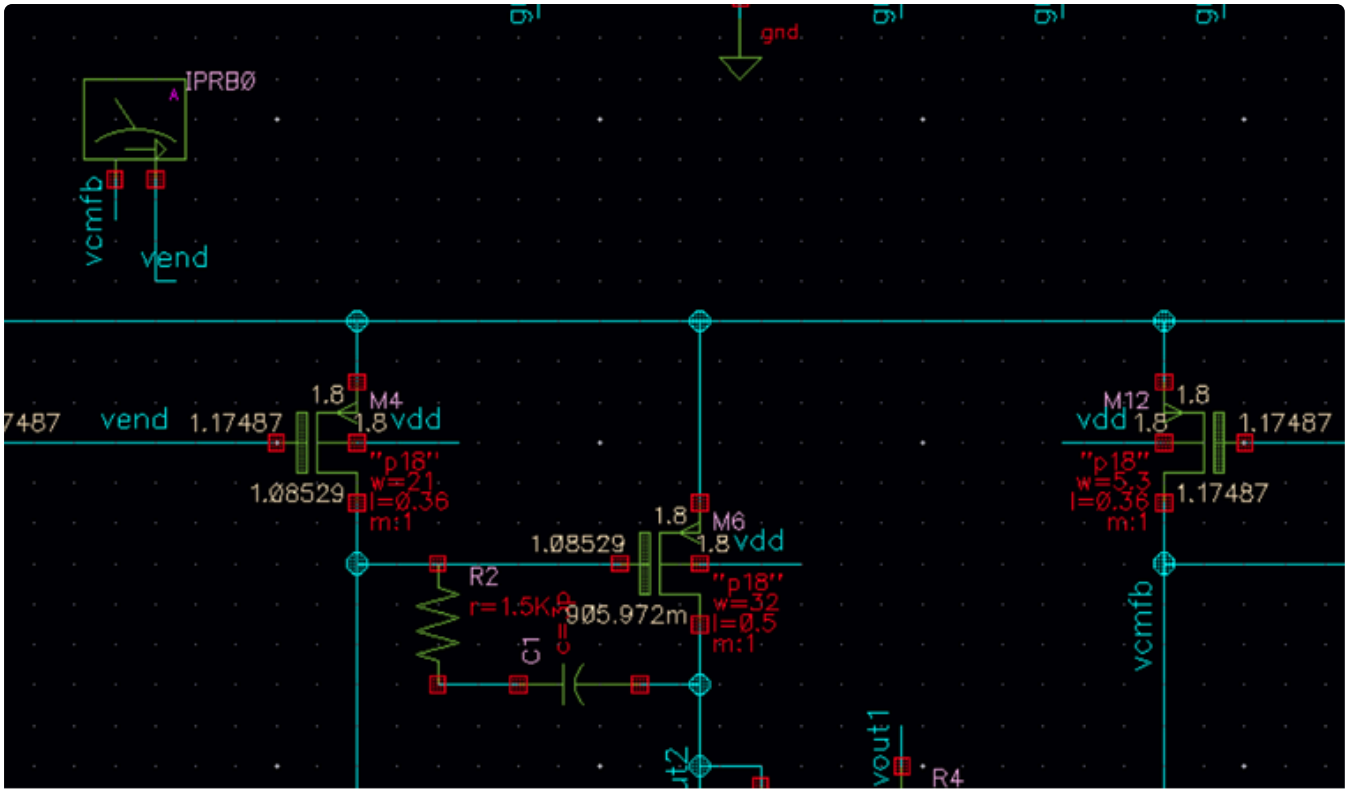


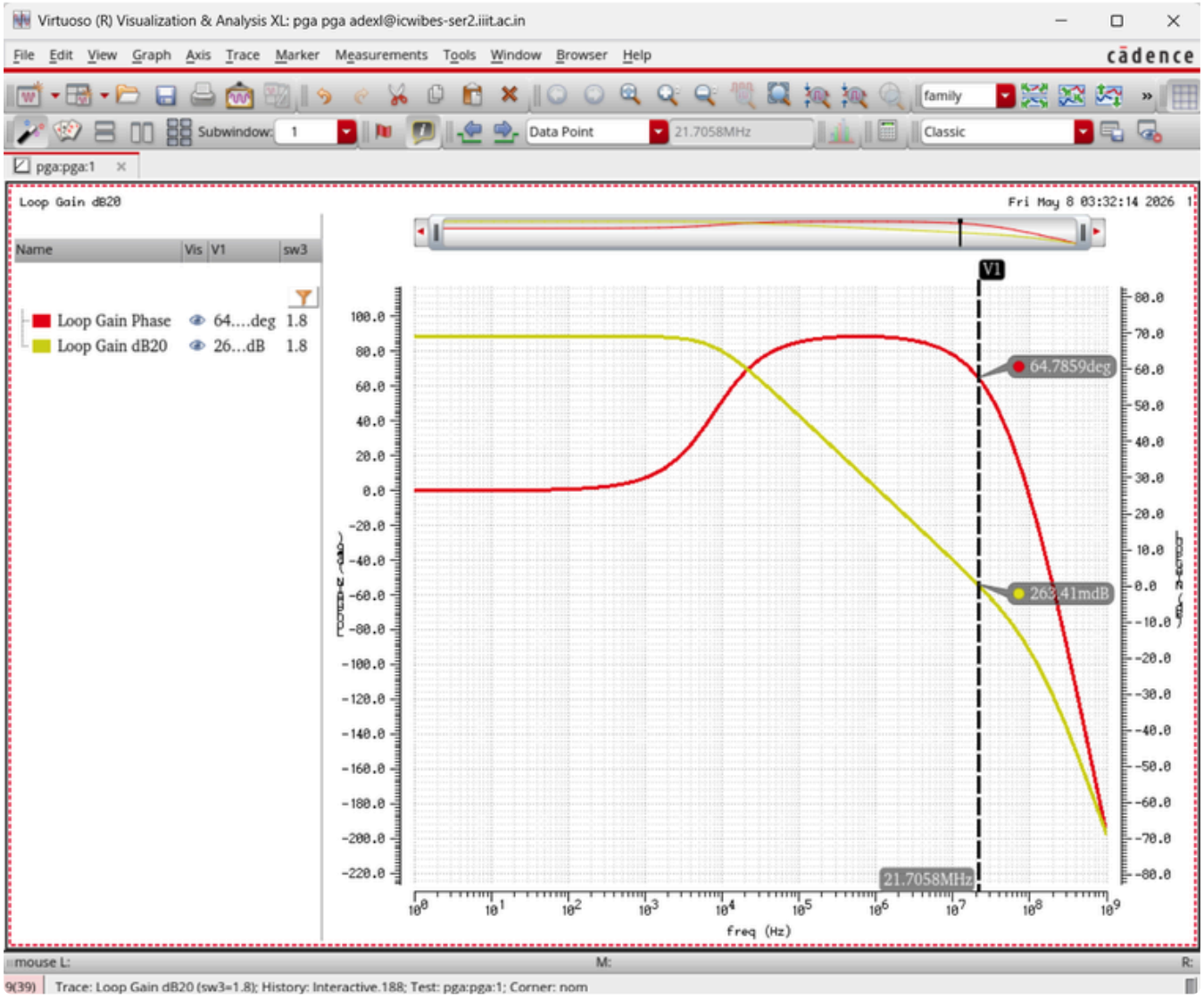


Loop	Result
Differential feedback loop	UGF 149 kHz, PM 92.9°, GM 96.6 dB at 462 MHz

Loop	Result
CMFB loop	Low-frequency gain ≈ 69 dB, UGF 21.7 MHz, PM 64.8°

The differential loop is unconditionally stable at this gain because the loop crosses over at 149 kHz, far below the OTA poles. The low-frequency loop gain is about 25 dB, which is A0 minus the 50 dB closed-loop gain (75.4 − 50).


gnd
IPRB0
vcmfb
vend
M4
1.8
1.8vdd
"p18"
w=21
l=0.36
m:1
1.08529
1.17487
M6
1.8vdd
w=32
l=0.5
R2
r=1.5K
905.972m
C1
M12
vdd 1.8
w=5.3
vout1
R4


Virtuoso (R) Visualization & Analysis XL: pga pga adexl@icwibes-ser2.iiit.ac.in
File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help
cadence
Subwindow:
Data Point
21.7058MHz
Classic
pga:pga:1
Loop Gain dB20
Fri May 8 03:32:14 2026
Name
Vis
V1
sw3
Loop Gain Phase
64....deg
1.8
Loop Gain dB20
26...dB
1.8
64.7859deg
21.7058MHz
Freq (Hz)
mouse L:
M:
R:
Trace: Loop Gain dB20 (sw3=1.8); History: Interactive.188; Test: pga:pga:1; Corner: nom

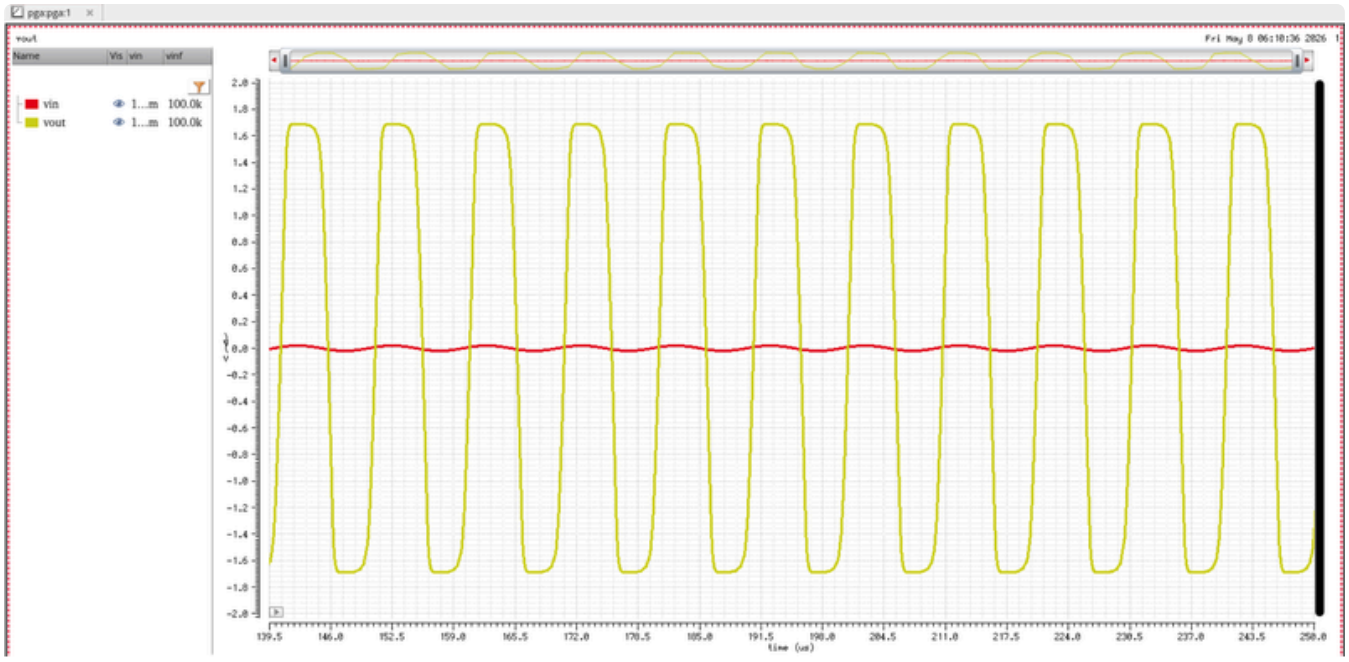
7.4 Programmable gain

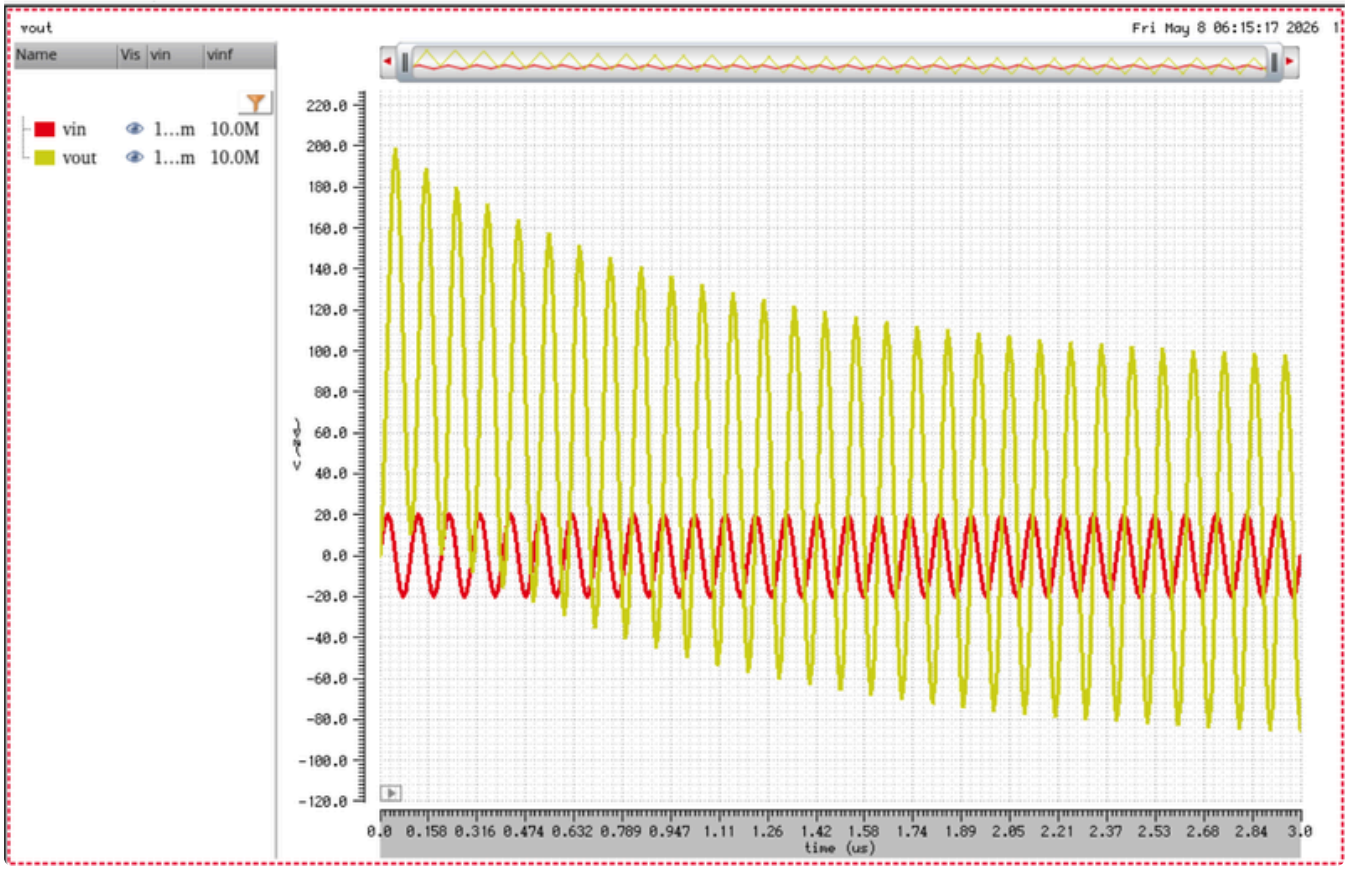
Code	Ideal (V/V)	Simulated (dB)	Simulated (V/V)	Gain error	−3 dB frequency
50 dB	316	49.53	298.4	−5.6%	153 kHz
36.7 dB	68.3	36.58	67.5	−1.2%	712 kHz
23.3 dB	14.6	23.29	14.6	0.0%	3.13 MHz
10 dB	3.16	10.13	3.21	+1.6%	11.5 MHz

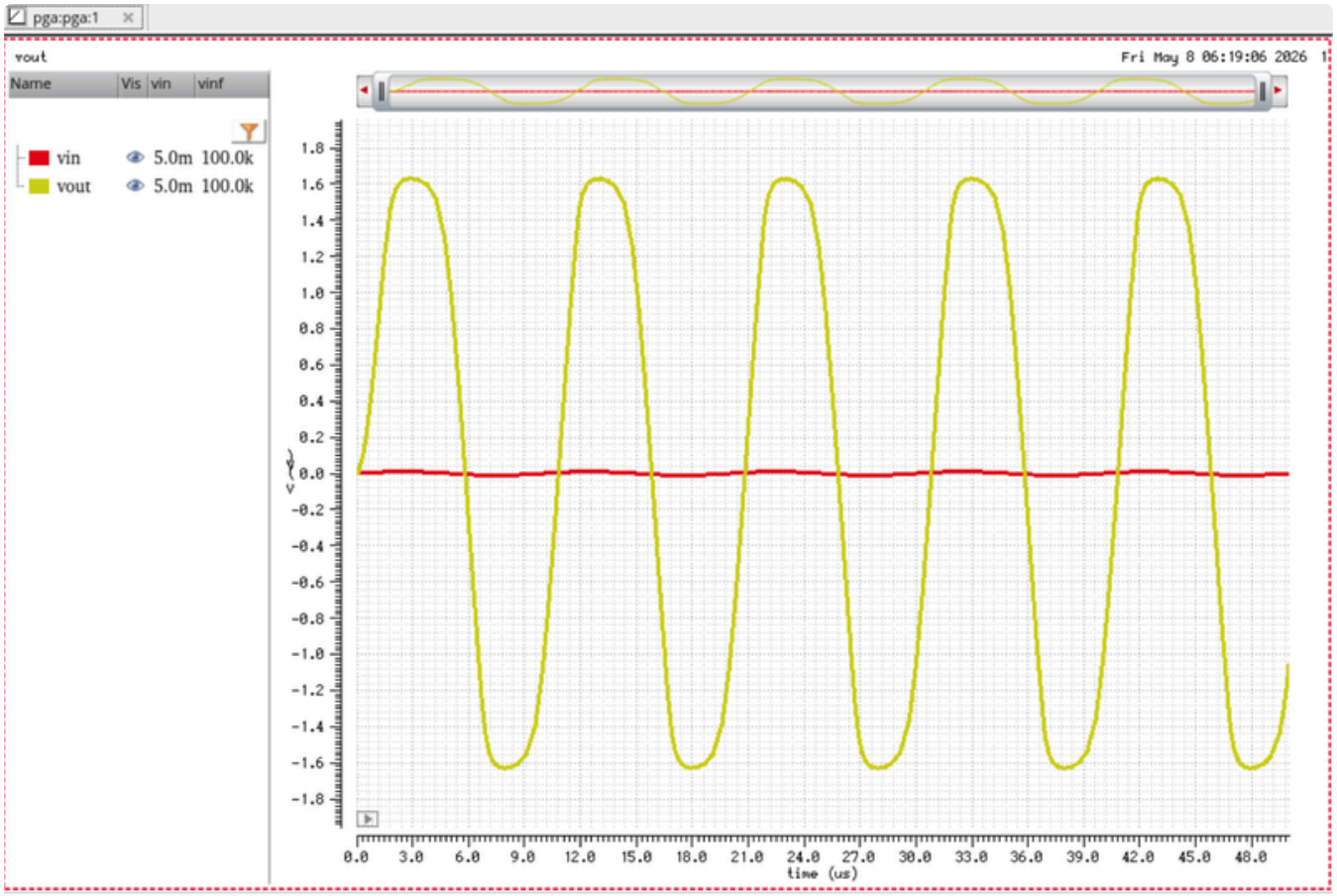
Gain × bandwidth stays near 46–48 MHz for the first three codes (298 × 153 kHz = 45.7 MHz), which confirms a single-pole closed-loop response set by the OTA. At 10 dB the product drops to 37 MHz because the second pole starts to matter. The 50 dB gain error of 5.6% follows from finite loop gain: with T0 ≈ 18.6 the error is 1 / (1 + T0) ≈ 5%, and switch resistance adds the rest.

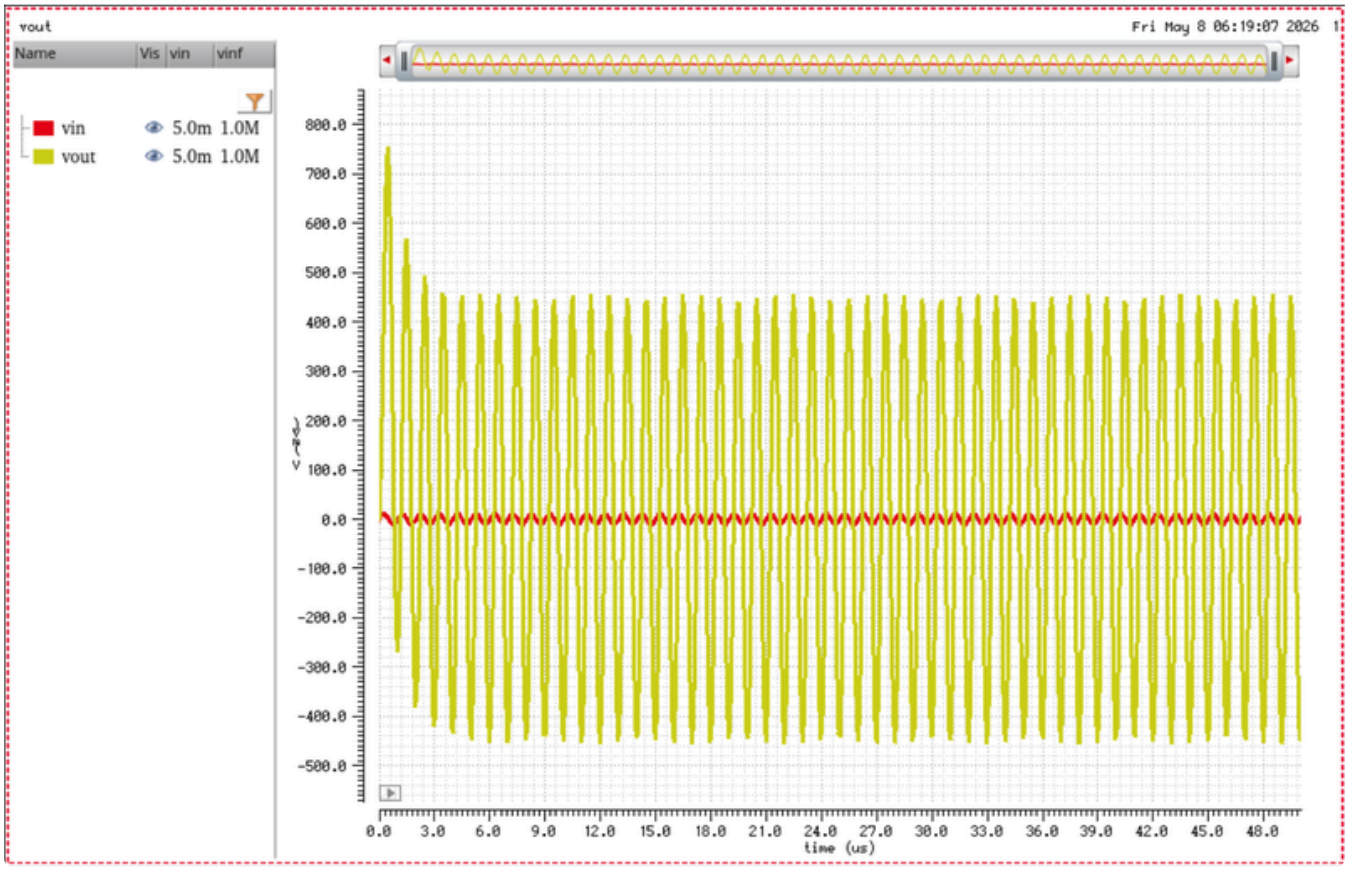
7.5 Transient response

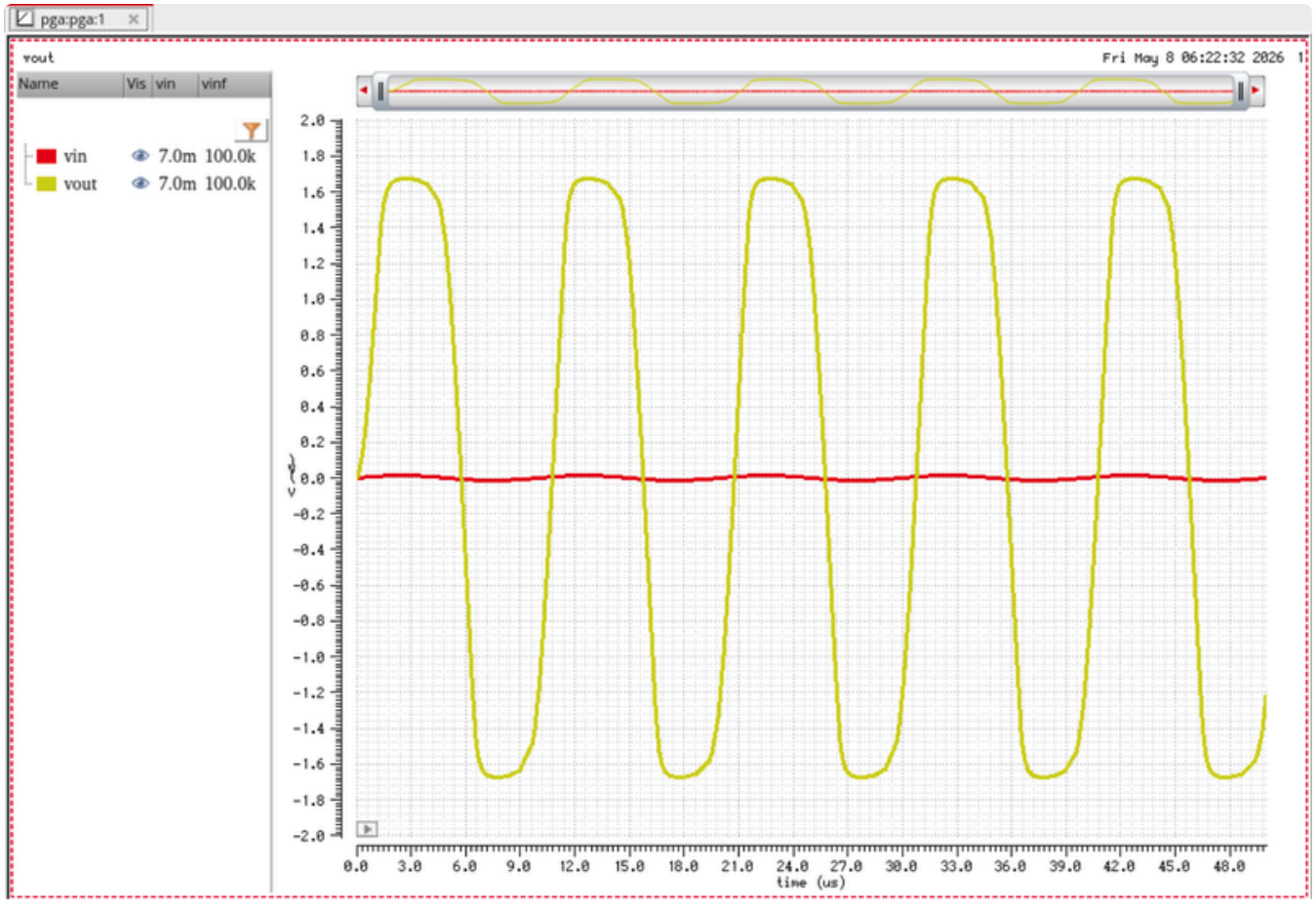
At the 50 dB code the output clips near ±1.7 V for inputs above about 5 mV peak, so the linear input range is about 5 mV. The 10 mV, 100 kHz case clips; the 10 MHz case is far above the 153 kHz bandwidth and the output is strongly attenuated.

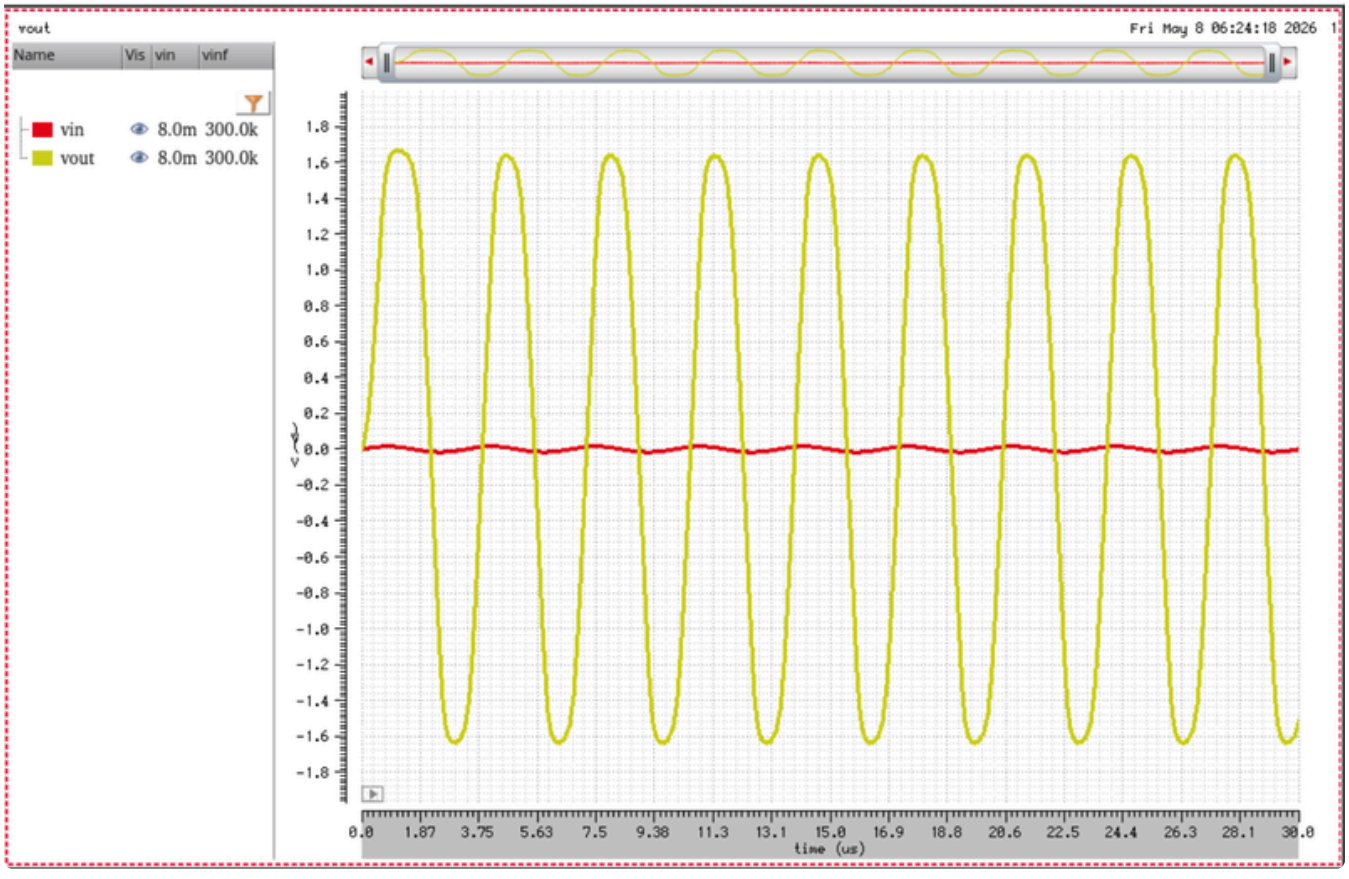

vout
Name
Vis
vin
vinf
vin
vout
1...m
100.0k
Fri May 8 06:10:36 2026
time (us)


vout
Fri May 8 06:15:17 2026
Name
Vis
vin
vinf
vin
vout
1...m
10.0M
time (us)


pga:pga:1
vout
Fri May 8 06:19:06 2026
Name
Vis
vin
vinf
vin
5.0m
100.0k
vout
5.0m
100.0k
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
-0.2
-0.4
-0.6
-0.8
-1.0
-1.2
-1.4
-1.6
-1.8
0.0
3.0
6.0
9.0
12.0
15.0
18.0
21.0
24.0
27.0
30.0
33.0
36.0
39.0
42.0
45.0
48.0
time (us)


vout
Fri May 8 06:19:07 2026
Name
Vis
vin
vinf
vin
5.0m
1.0M
vout
5.0m
1.0M
800.0
700.0
600.0
500.0
400.0
300.0
200.0
100.0
0.0
-100.0
-200.0
-300.0
-400.0
-500.0
0.0
3.0
6.0
9.0
12.0
15.0
18.0
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42.0
45.0
48.0
time (us)

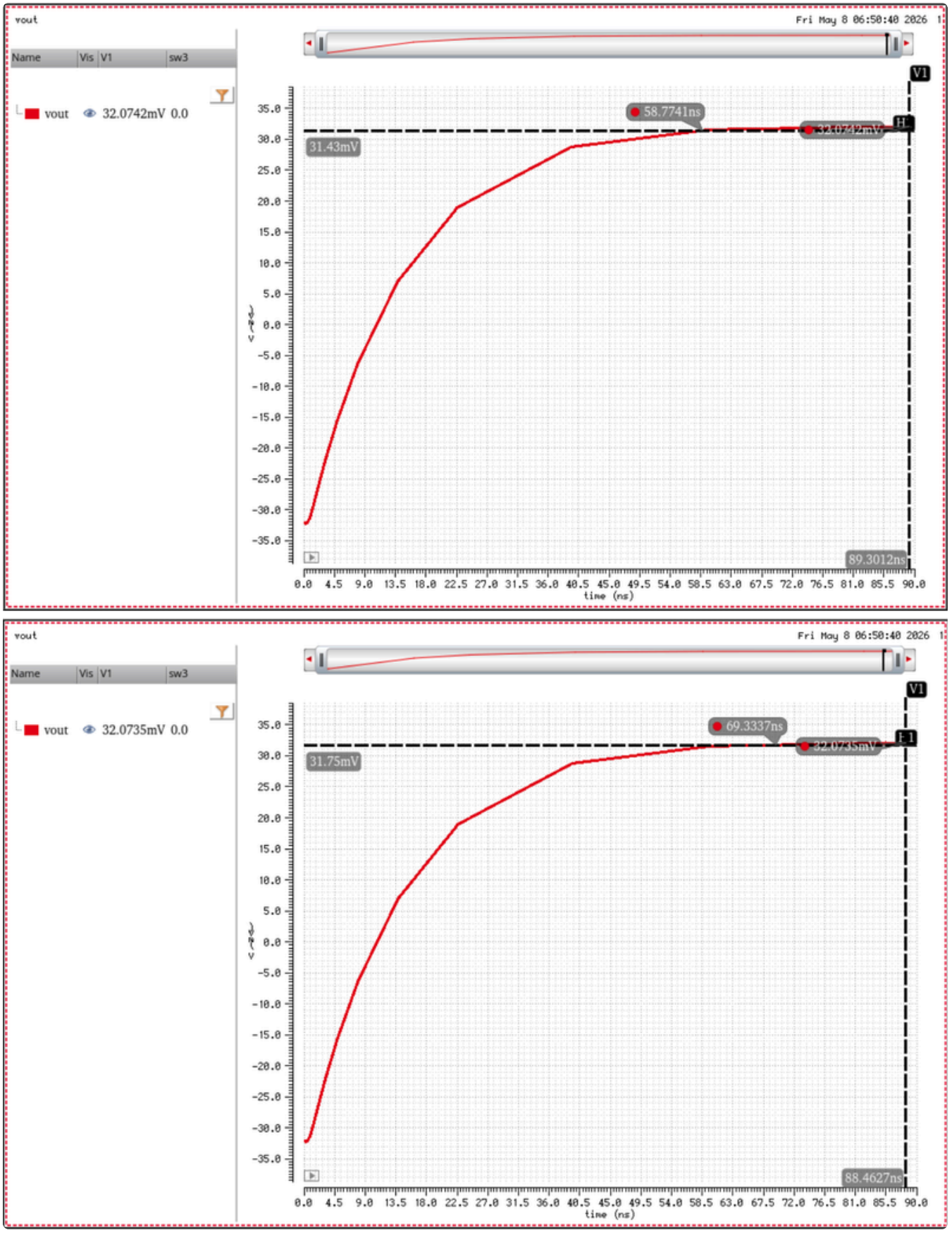




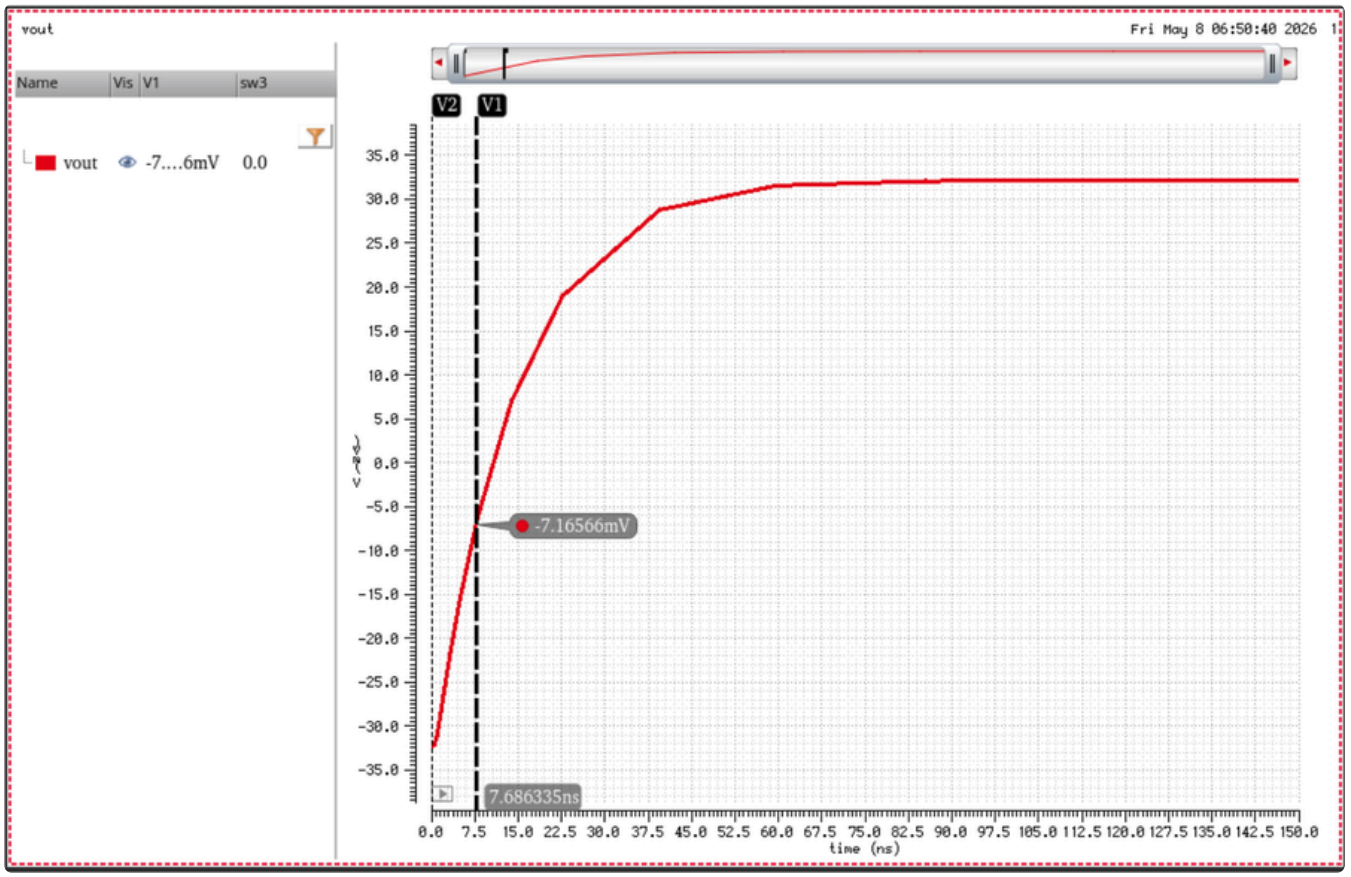
Settling was measured at the 10 dB code for a 20 mV input step, which gives a 64 mV output step.

Band	Settling time
2%	58.8 ns
1%	69.3 ns

There is no overshoot between the two bands, and both are well inside the 200 ns target.

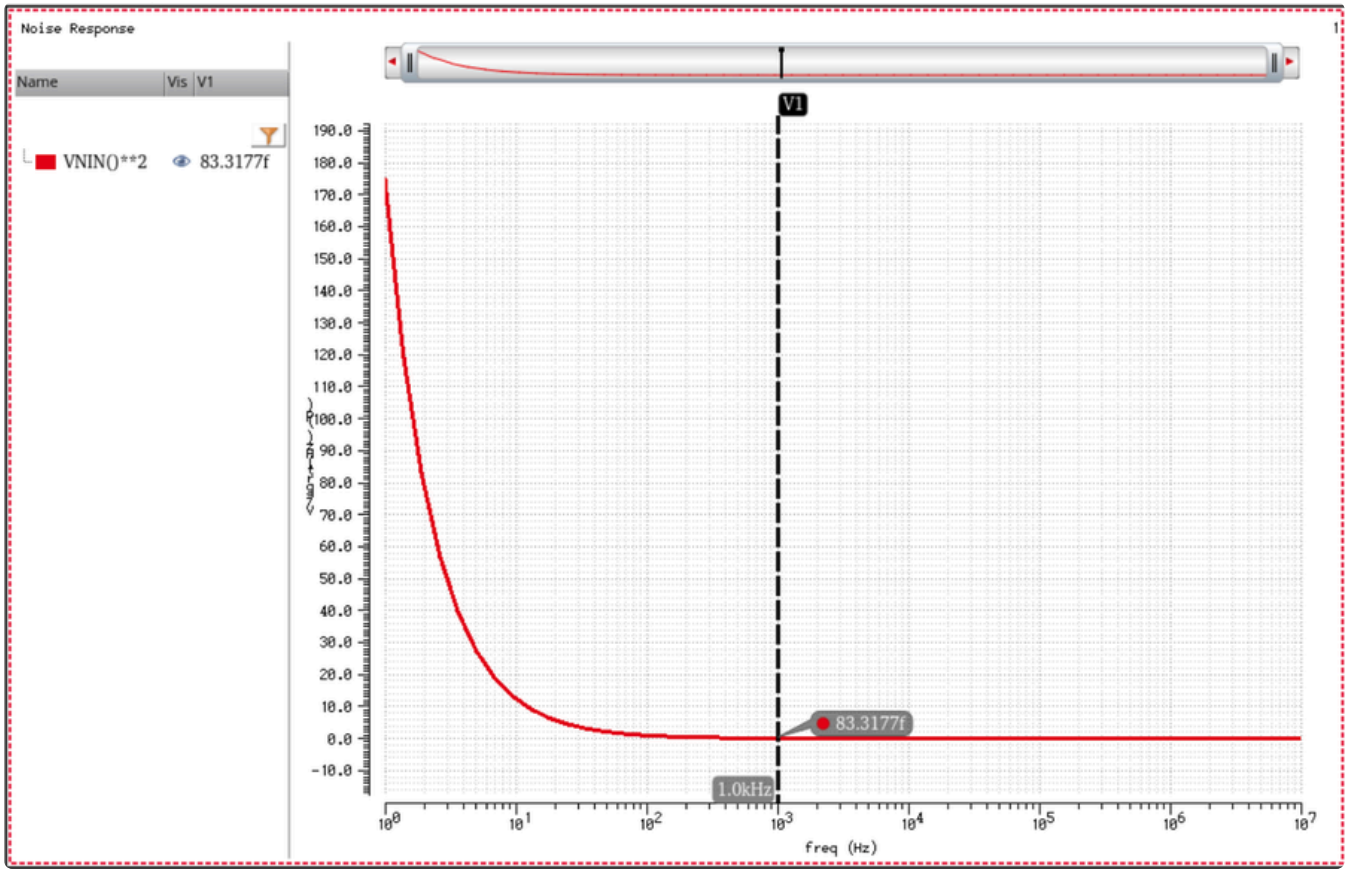


The closed-loop slew measurement at the 10 dB code gives 24.8 mV in 7.69 ns, or 3.23 V/µs. That is a small-signal rate limited by the closed-loop bandwidth, not the OTA slew rate. The OTA slew rate Itail / Cc = 50 V/µs is a hand estimate.



7.6 Noise

The plot shows input-referred noise power density falling from a 1/f region below about 100 Hz to a flat floor. At 1 kHz the marker reads 83.3 fV²/Hz as plotted, which is about 289 nV/√Hz.



8. Summary and Conclusion

The PGA covers 10–50 dB with 75.4 dB OTA gain, a stable differential loop and 833 µW OTA power. Results against each target are below, followed by a discussion of the bandwidth, noise and CMFB figures.

Parameter	Target	Result
DC open-loop gain	≥ 70 dB	75.4 dB
Open-loop −3 dB bandwidth	≥ 75 kHz	about 7–8 kHz (UGB / A0)
Unity-gain bandwidth	—	42.5 MHz
Phase margin	60°–80°	85.6°
CMRR	≥ 80 dB	150–176 dB (ideal, matched)
Slew rate	≥ 20 V/µs	50 V/µs (hand estimate)
Settling time (2%)	≤ 200 ns	58.8 ns
Power (OTA + CMFB)	≤ 1 mW	833 µW
Input noise at 1 kHz	≤ 35 nV/√Hz	about 289 nV/√Hz

Parameter	Target	Result
Output offset (DCOC off)	⩽ 10 mV after DCOC	5.97 mV
CMFB gain / UGF / PM	⩾ 40 dB / 5–10 MHz / ⩾ 60°	69 dB / 21.7 MHz / 64.8°
PGA gain range	10–50 dB	10.1–49.5 dB
PGA bandwidth at 50 dB	⩾ 750 kHz	153 kHz
PGA gain error	⩽ 5%	0–1.6% at three codes, 5.6% at 50 dB

Discussion

- **Bandwidth.** The 750 kHz bandwidth at 50 dB needs a 237 MHz UGB. With the power-limited UGB (53 MHz by hand, 46 MHz simulated) the 50 dB code has a 153 kHz bandwidth, the 36.7 dB code reaches 712 kHz, and the lower codes reach several MHz. The open-loop −3 dB frequency follows UGB / A0 for the same reason.
- **Gain error.** At 50 dB the loop gain is about 18.6, which sets the error near 5%; the error falls at lower codes, where the loop gain is higher.
- **CMFB.** The loop has 69 dB gain and 64.8° phase margin. The 21.7 MHz UGF can be lowered by adding capacitance on the control node.
- **Noise.** The 1 kHz density lies in the 1/f region of the input pair. A larger input-pair area and lower DCOC resistor noise would reduce it.

Conclusion

The design delivers a stable, programmable 10–50 dB PGA that meets the gain, settling and OTA power targets. The bandwidth at the highest gain code is set by the OTA unity-gain bandwidth, which the 1 mW budget limits to about 46 MHz.

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